Endometriosis

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Microbiome Signatures, Microbiome Medicine

Abstract. Endometriosis affects 11% of women. Exploring the Microbiome Signature of endometriosis may lead to better treatments and understanding.

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Overview

Endometriosis affects about 10-15% of women globally, causing chronic pelvic pain, infertility, and other severe symptoms. [1] While experts often point to retrograde menstruation as the cause of endometriosis, this theory falls short because it fails to explain why many women with retrograde menstruation never develop endometriosis, nor does it account for cases in prepubertal girls, postmenopausal women, *men*, or endometriotic lesions found in locations far from the pelvis, such as the lungs. [2] This clearly indicates that other significant pathophysiological factors are involved. The lack of a definitive cure and the suboptimal effects of current treatments highlight the urgent need for a breakthrough.

Associated Conditions

While endometriosis primarily affects the endometrium, it can have systemic effects and influence various aspects of a person's health. Individuals with endometriosis may be at an increased risk for other conditions. These include chronic pelvic pain, infertility, irritable bowel syndrome (IBS), interstitial cystitis, autoimmune conditions, and ovarian cancer. [3][4] Researchers have also documented that many of these comorbidities involve urogenital or gut dysbiosis, suggesting potential microbiome signature overlaps between the conditions. [5]

Causes

Prevailing causal theories of endometriosis—such as retrograde menstruation, Müllerianosis, and lymphovascular metastasis—provide foundational insights but fall short of fully explaining the condition's complexity. A more comprehensive, systems-based approach that integrates these theories with emerging insights—such as the role of the microbiome and the microbial metallomics theory of endometriosis—offers a more nuanced and integrated understanding of the disease. By challenging entrenched assumptions and embracing granular, data-driven perspectives, we can identify previously overlooked mechanisms driving pathogenesis. This shift not only enriches our understanding of endometriosis but also paves the way for highly personalized treatments that leverage microbiome research, dramatically improving patient outcomes and redefining the standard of care.

▶ **What are the current causal theories of endometriosis and their limitations?**

Diagnosis

Laparoscopy remains the gold standard for diagnosing endometriosis. Despite its accuracy, laparoscopy is still an invasive procedure with associated risks and costs, necessitating the development of less invasive diagnostic tools. As researchers and clinicians increasingly leverage microbiome signatures and metabolomic signatures for non-invasive diagnostic purposes and treatment guidance, these tools are more likely to become mainstream in medical practice.[17][18] Their growing utility in providing earlier, more accurate diagnoses and tailoring targeted therapies underscores their potential to revolutionize non-invasive diagnostics, particularly in conditions like endometriosis.[19]

Primer

The interplay between metal homeostasis, estrogen metabolism, and metabolomic disruptions in endometriosis reveals how systemic and localized factors converge to drive endometriosis progression. These interrelationships underscore the microbiome's central role in the disease's pathogenesis and highlight opportunities for microbiome-based diagnostics and microbiome-targeted interventions (MBTIs).

Nutritional Immunity

Several nutritional immunity factors are found to be elevated in endometriosis. Studies suggest that levels of lactoferrin are significantly increased in individuals with endometriosis, reflecting a heightened immune response. [20] The increased levels of these proteins indicate an attempt by the body to sequester nutrients and metals from pathogens, while also modulating the inflammatory environment associated with endometriosis. This suggests that understanding the role of nutritional immunity could provide insights into the management and treatment of endometriosis.

Metallomic Signature

High iron levels from retrograde menstruation promote pathogenic bacteria, lesion growth, inflammation, and oxidative stress.[21] Studies suggest that nickel sensitivity affects 90.3% of patients, or possibly even more [22], and that a Low-nickel diet (LNiD) shows significant clinical benefits. [23] Zinc, despite its antioxidative and anti-inflammatory role, has been significantly associated with an increased risk of endometriosis owing to the activity of matrix metalloproteinases (MMPs), which are implicated in endometriotic lesion invasion.[24]Metalloestrogens like cadmium, nickel, and lead increase oxidative stress, inflammation, and estrogenic activity. [25] These findings highlight the role that the metallomic signature plays in the condition, and the potential of chelation therapies, dietary modifications, and reduced heavy metal exposures for managing or preventing endometriosis.

▶ **What is the metallomic signature of endometriosis?**

Estrogen Metabolism

The estrobolome, a subset of gut microbiome genes, regulates estrogen metabolism and plays a critical role in endometriosis progression. Major microbial associations (MMAs) in the endometriosis microbiome signature, such as *Escherichia coli*, *Bacteroides fragilis*, and *Streptococcus agalactiae* produce β-

glucuronidase, increasing estrogen deconjugation and circulating hormone levels. This microbial activity drives inflammation, angiogenesis, hormonal dysregulation, and cellular changes, contributing to endometriosis pathogenesis. Dysbiosis in endometriosis is both a cause and consequence of the disease. [41][42][43]

Interventions

Our validation method confirms microbiome-targeted interventions (MBTIs) by aligning therapeutic effects on microbial imbalances with clinical outcomes, key pathological markers, and major microbial associations (MMAs) of endometriosis. This dual approach ensures that interventions address both dysbiosis and the condition's core biological mechanisms, reinforcing the accuracy of microbiome signatures and the efficacy of MBTIs. Promising candidates with partial alignment or preliminary benefits are also identified for further research, ensuring a pipeline of innovative therapies for future validation.

Intervention	Classification	MBTI Status
Pueraria flower extract (PFE)	Herbal	Validated
Metronidazole	Pharmaceutical	Validated
Metformin	Drug Repurposing	Validated
Low-Nickel Diet	Diet	Validated
Hyperbaric Oxygen Therapy (HBOT)	Biophysical Interventions	Validated
Microbiome Diet for Endometriosis	Diet	Promising Candidate
Lactoferrin	Supplement	

▶ **What interventions are still in the process of validation as an MBTI?**

STOPs

A STOP (Suggested Termination Of Practice) recommends discontinuing medical interventions shown to be ineffective, harmful, or counterproductive based on emerging evidence. In endometriosis, heavy menstrual bleeding often leads to iron-deficiency anemia, traditionally treated with iron supplementation. However, iron supports bacteria linked to endometriosis, and iron supplementation is associated with increasing adverse effects. [x] Paradoxically, iron chelation therapies may improve absorption while reducing inflammation, highlighting the need to re-evaluate traditional practices. This illustrates how STOPs advance evidence-based care by addressing outdated or counterproductive treatments.

▶ **What STOPs should be further considered?**

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